

Operating Systems

BCA — Semester IV — 2024 — Answer Sheet

Subject Code: CACS251

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group C Attempt any TWO questions[2x10=20]

2. Define kernel. Explain different types of kernel in operating system.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What is PCB? What are different information that are included in a PCB?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. How paging can be used for memory management? Also differentiate it with segmentation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What are page faults and thrashing? Explain demand paging with protection bit.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. How deadlock can be prevented? Explain in detail.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Explain different file access methods in detail.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is distributed system? Explain the role of clock synchronization in distributed system.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is critical section problem? How semaphores can be used to solve critical section problem Also explain producer consumer problem with its solution using Mutex.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is Belady's Anomaly in page replacement Algorithm? Explain with example. Consider the following page reference string: 7, 2, 3, 1, 2, 5, 3, 4, 6, 7, 7, 1, 0, 5, 4, 6, 2, 3, 0, 1. How many page faults will occur if the program has four page frames available to it and uses the following page replacement algorithm?
a) FIFO replacement
b) LRU replacement
c) Optimal replacement

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is the use of DMA controller? Consider the following process and answer the following questions.

Process	Allocation	Max Available																																								
A	B	C	D																																							
P0	0	1	10	0	1	2	1	5	2	0	P1	1	0	0	1	7	5	0	P2	1	3	5	4	2	3	5	6	P3	0	6	3	20	6	5	2	P4	0	1	40	6	5	6

a) What is the content of need matrix?
b) Is the system in safe state?
c) If P2 request (3 4 0 2), can the request be granted immediately.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.